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Assessment 15

Review Learning Objectives

coach

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Assignment details

Due
Dec 27, 11:59 PM IST

Attempts
Unlimited

Retry

Submitted
Dec 11, 4:04 AM IST

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1. Which of the following are essential components in the quantum teleportation protocol? **0.6 / 1 point**

- [x] Quantum entanglement generator.

  This should not be selected
  No, while entanglement is necessary, there's no specific 'entanglement generator' used in the protocol.

- [ ] Measurement in the Bell basis.
- [x] Entanglement between two qubits.

  Correct
  Correct! Entanglement is a core component of the quantum teleportation protocol.

- [x] A classical communication channel.

  Correct
  Correct! Classical communication is required to complete the teleportation process.

- [ ] Quantum cloning machine.

2. Why is the no cloning theorem important in quantum mechanics? **1 / 1 point**

- ( ) It helps increase the power of classical computers.
- ( ) It enables the creation of quantum entanglement.
- ( ) It allows for faster-than-light communication.
- (•) It ensures that quantum information cannot be perfectly copied.

  Correct
  Correct! The no cloning theorem is fundamental as it states that an arbitrary quantum state cannot be copied perfectly.

3. What is the primary advantage of quantum key distribution (QKD) over traditional encryption methods? **1 / 1 point**

- ( ) Faster encryption speed
- ( ) Lower cost of implementation
- (•) Immunity to computational attacks
- ( ) Ease of use

  Correct
  Correct! QKD offers immunity to computational attacks because it relies on quantum mechanics.

4. What are the necessary components for generating and transferring entangled photons? **0.6666666666666666 / 1 point**

- [x] Photon source

  Correct
  Correct! A photon source is necessary to generate photons for entanglement.

- [x] Optical fiber

  Correct
  Correct! Optical fibers are often used to transfer entangled photons over distances.

- [ ] Classical channel
- [x] Beam splitter

  Correct
  Correct! A beam splitter is used to create entangled photon pairs.

- [x] Quantum repeater

  This should not be selected
  Quantum repeaters are used to extend the range of quantum communication, but they are not necessary for the initial generation and transfer of entangled photons.

- [ ] Polarizer

5. Which concept is essential for explaining quantum teleportation? **1 / 1 point**

- ( ) Photon polarization
- ( ) Classical communication
- (•) Quantum entanglement
- ( ) Quantum decoherence

  Correct
  Correct! Quantum teleportation relies on the phenomenon of quantum entanglement.